Vishnu Vardhan

Forward Deployed Engineer

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Experience

The Developers Arena — Bengaluru, Karnataka, India

Software Engineer Intern – Client-Facing AI & Backend Systems — Nov 2025 – May 2026

- **Problem:** client integrations were bottlenecked by slow, inconsistent backend APIs across multiple live deployments. Diagnosed latency at the API layer and **architected 6+ production-grade REST APIs** with **Flask and FastAPI** using **modular MVC design**, cutting average response latency by **30%** across client environments.
- Partnered on client requirements for **document Q&A, semantic search, and automated reasoning** and delivered the fix as **end-to-end AI backend pipelines — OpenAI/Gemini LLMs, LangChain agents, and RAG** on **Pinecone/ChromaDB** — shipped directly to production.
- Root-caused slow query performance under load and **re-engineered MySQL schemas** with normalized design, composite indexing, and stored procedures — cutting query execution time by **40%** and holding up under high-volume concurrent workloads.
- Diagnosed a throughput ceiling during a client scale-up and closed it by introducing **Apache Kafka event streaming, Redis caching, and Celery async task queues** — **3× API throughput**, sustaining **sub-200ms p95** under **500+ concurrent requests**.
- Owned security hardening end-to-end ahead of a client compliance review: **JWT/OAuth2, RBAC, AES-256 encryption, parameterized queries** — delivered **zero critical CVEs** across all API-facing and blockchain endpoints.
- Diagnosed deployment downtime as an architectural issue, not a config one, and **re-architected the monolith into Dockerized microservices** with **Kafka messaging and Web3.py audit trails** — **50% less downtime**, enabling **zero-downtime rolling updates**.
- Operated as the connective layer across backend, AI/ML, blockchain, QA, and DevOps squads — sustained **95%+ sprint velocity** across **8+ Agile cycles**. **View Certificate**

Infotact Solutions — Bengaluru, Karnataka, India

Software Engineer Intern – Agentic AI Systems — Sep 2025 – Dec 2025

- **Problem:** content research and drafting were manual, slow, single-threaded processes. **Architected a multi-agent research and content pipeline** in **Python (LangChain, LangGraph)** with modular agent roles for web research, summarization, and drafting inside a stateful workflow.
- Built **autonomous multi-step reasoning** via **LangGraph state machines** with tool-calling, agent-to-agent delegation, and short/long-term memory — letting agents plan, execute, and self-correct without hand-holding.
- Exposed the workflow to real usage by integrating **OpenAI/Claude LLM APIs** with an **async FastAPI backend** for triggering, monitoring, and retrieving results in real time.
- Diagnosed hallucination as the core reliability risk and closed the gap with a **RAG pipeline** on **Pinecone/ChromaDB** and custom embedding pipelines — grounded outputs in source data, improving factual accuracy.
- Found and fixed a latency bottleneck under multi-agent load using **Python asyncio orchestration, response caching, and parallelized agent calls** — reduced end-to-end pipeline latency. **View Certificate**

Skills

Programming Languages: Python,
Backend & APIs: Flask, FastAPI, REST API Design
Databases: MySQL, PostgreSQL (pgvector), MongoDB, Vector Databases (Pinecone, ChromaDB)
Agentic AI Orchestration: LangChain, LangGraph, MCP (Model Context Protocol), Multi-Agent Systems, Multi-LLM Routing & Fallback Strategies
AI / LLM: LLM Integration (OpenAI GPT-4, Gemini, Claude, Groq), RAG (retrieval, chunking, re-ranking, hybrid search, caching), Prompt Engineering, Vector Embeddings, Hallucination Prevention, Evaluation & Observability
Enterprise AI Platforms : AWS Bedrock, Azure OpenAI, Google Vertex AI concepts
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD Pipelines, Kafka, Redis, Celery
Adjacent Skills: Root-cause diagnosis, client-facing requirements gathering, production deployment ownership, cost/latency benchmarking
Tools & Platforms: Git, GitHub, Postman, VS Code, Jira, n8n, Jupyter Notebook, Vercel, Netlify

Projects

Zero-Hallucination RAG Intelligence Platform | *React (Vite), Python, FastAPI, PostgreSQL (pgvector), OpenAI GPT-4, RAG, LLM*

- **Business problem:** standard LLM tools hallucinate on document-specific queries, producing confident but unverifiable answers with no traceability to source evidence.
- **Architecture:** architected and shipped a full-stack **Retrieval-Augmented Generation** platform integrating the **OpenAI GPT-4 API** with **hybrid dense + lexical search** over **PostgreSQL (pgvector)**, using **TanStack Start + React** for a server-rendered, production-grade frontend.
- Engineered a **6-layer hallucination-prevention system** — retrieval-confidence thresholds, clause-affinity re-ranking, evidence-overlap validation, prompt-contract enforcement, and post-generation **claim verification** — so the model **abstains rather than fabricates** when evidence is weak, missing, or conflicting.
- Built **multi-format ingestion** (PDF, DOCX, XLSX, CSV, TXT, Markdown, JSON, HTML) processing **421 indexed chunks across 3 live documents**, with automated chunking, embedding, and vector indexing delivering **traceable inline citations** on every generated claim.
- **Result:** benchmarked **0.0% hallucination rate** and **1.000 groundedness** across **40 evaluation queries** (24 answered, 16 correctly abstained) — with **1.000 recall@K**, **0.797 precision@K**, **1.000 MRR**, and **0.681 faithfulness**, at **2.3s average end-to-end latency**.
- **Next steps scoped:** deeper **database and third-party API integrations** to expand structured data grounding — pushing faithfulness and precision higher while extending **zero-hallucination** answers to live relational data. **Live App**

N45 Agentic Builder | *Internship Project — Next.js, React, TypeScript, FastAPI, Python, MySQL, LLM, LangGraph, LangChain, RAG, Vector DB, AI*

- **Business problem:** enterprises lacked a unified visual platform to build, orchestrate, and monitor multi-agent AI workflows — relying instead on fragmented scripts and manual integrations.
- **Architecture:** contributed to a production-ready **Enterprise AI Agent Development Platform** featuring a drag-and-drop **Workflow Builder (React Flow)** and **multi-agent orchestration** for parent–sub-agent collaboration.
- Worked on **workflow validation, versioning, autosave, deployment, and real-time execution monitoring** for reliable end-to-end workflow lifecycle management.
- Helped integrate **multiple LLM providers** (**OpenAI, Gemini, Anthropic, Groq, Ollama, Azure OpenAI**) via configurable AI profiles, with **Knowledge Base (RAG)** support for document ingestion and semantic retrieval.
- Supported implementation of **JWT authentication, RBAC, multi-tenant workspace architecture, and encrypted credential management** for enterprise-grade security.
- Assisted in connecting enterprise tools (**Google Workspace, Slack, GitHub, Jira, Notion, REST/GraphQL APIs, webhooks**) with **analytics dashboards and audit logging** for full observability.
- **Result:** platform enabled enterprises to build and scale AI agent workflows without custom infrastructure — cutting integration overhead across LLM providers and tools. **Live App**

Agentic HR – AI-Powered Recruitment Pipeline | *Python, LangGraph, FastAPI, Groq LLM, Twilio, Google Calendar API, Gmail API, MongoDB*

- **Business problem:** manual recruitment coordination across shortlisting, screening, scheduling, and onboarding was slow and error-prone for HR teams.
- **Root cause & architecture:** no single system owned the full candidate lifecycle. Designed an **8-node multi-agent pipeline** on **LangGraph StateGraph**, automating end-to-end from resume screening to onboarding with **3 human-in-the-loop approval gates** for client trust.
- Shipped a **Resume Shortlister Agent** (**Groq LLM, Llama 3.3 70B**) parsing PDF/DOCX resumes and ranking the top **15 candidates** against job descriptions with match scores and rationale.
- Shipped a **Pre-Screening Call Agent** (**Twilio voice + Edge TTS**) running autonomous live phone screens, extracting **5 structured data points** per candidate (CTC, availability, job-change intent).
- Shipped an **Interview Scheduler Agent** (**Google Calendar + Gmail APIs**) that scans availability, books the first free 1-hour slot, and generates Meet links with zero manual coordination.
- Shipped an **Onboarding Agent** that auto-sends personalized welcome emails, closing the loop on the full hire-to-onboard workflow.
- **Result & next steps:** identified production upgrade path — migrating voice to low-latency streaming to remove turn-based delay, and scoping a **Background Verification Agent** for post-onboarding checks. **GitHub Repo**

Education

Aditya Degree College — B. Kotha Kota

Bachelor of Computer Applications (BCA); CGPA: 9.02/10 — 2026